



COLLEGE EXAMINATIONS – ACADEMIC YEAR 2024- 2025

END OF SEMESTER ONE FINAL EXAMINATION

MODULE CODE & TITLE: MAT1264 ADVANCED CALCULUS FOR ENGINEERS

SCHOOL: ALL CST SCHOOLS

DEPARTMENT & PROGRAMMES: ALL CST DEPARTMENTS &

PROGRAMMES

FIRST YEARS: CIV ENG, SGE, TRE, WREE, ENE, and MEE

SEMESTER 2

DATE: 02/June/2025

TIME: 2 hours

MAXIMUM MARKS = 50

INSTRUCTIONS:

1. This paper contains **TWO** sections.
 2. Section A is compulsory, and Answer any **TWO** of the **THREE** questions in Section B.
 3. No written materials allowed into the Examination Room.
 4. Write all your answers in the answer booklet provided.
 5. Do not forget to write your Registration Number.
 6. Do not write any answers on this question paper.
 7. Where appropriate draw large clearly labeled diagrams in your answers.
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SECTION A. ATTEMPT ALL QUESTIONS /20 marks

Question one.

- a) Find the Wronskian of the set $\{x, \cos(-x), 1\}$ and state if the set of solutions is independent or not (2 marks)
- b) The Laplace transform of a function $f(t)$ is given as $F(s) = \frac{2s+1}{s(s+1)}$. State the Laplace transform of $e^{-2t}f(t)$. (3 marks)
- c) Find the $f(t)$ such that $\mathcal{L}\{f(t)\}$ is given as $F(s) = \frac{2}{s^2 + 4s + 5}$. (2 marks)
- d) By Laplace transforms solve the differential equation (3 marks)
- $$y'' - y = -t^2, \quad y(0) = 2, \quad y'(0) = 0.$$
- e) If $\phi(x, y, z) = x^2yz$ and $\vec{A} = 3x^2y\vec{i} + yz^2\vec{j} - xz\vec{k}$, then find $\frac{\partial^2}{\partial y \partial z}(\phi \vec{A})$ at the point $(1, -2, -1)$. (4 marks)
- f) Evaluate: $\int_{x=0}^1 \int_{y=0}^1 \int_{z=0}^1 (x+y+z) dz dy dx$. (3 marks)
- g) Verify that $u(x, y) = xy + \frac{x}{y}$ is a solution of the equation (3 marks)

$$y \frac{\partial^2 u}{\partial y^2} + 2x \frac{\partial^2 u}{\partial x \partial y} = 2x$$

SECTION B: ATTEMPT ONLY TWO (2) QUESTIONS /30 marks

Question two.

- a) Evaluate $\int_C xy^4 ds$, C is the right half of the circle $x^2 + y^2 = 16$ rotated in the counter clockwise direction. (8 marks)
- b) Use the change of variables for evaluating the following double integral: (7 marks)

$$\iint_D \ln(x^2 + y^2) dA, \text{ where } D = \{(x, y) \text{ such that } x^2 + y^2 = e \text{ and } x^2 + y^2 = 4\}$$

Question three.

- a) Find the general solution of the following homogeneous linear system, using the vector- matrix methods, (9 marks)
- $$\begin{cases} x_1' = x_1 + x_2 - x_3 \\ x_2' = 2x_1 + 3x_2 - 4x_3 \\ x_3' = 4x_1 + x_2 - 4x_3 \end{cases}$$

- a) Evaluate $\iint_S u(x, y, z) ds$ where S is the surface of the paraboloid $z = 2 - (x^2 + y^2)$ above the xy -plane and $u(x, y, z)$ equal to 1 (6 marks)

Question four.

- a) Evaluate $\int_{(1,2)}^{(3,4)} (6xy^2 - y^3)dx + (6x^2y - 3xy^2)dy$ (7 marks)

b)

- i. Evaluate $\oint_C xydx + x^2 dy$ around the sides of the square with vertices $D(0,0), E(1,0), F(1,1)$ and $G(0,1)$. (5 marks)
- ii. Convert the line integral to a double integral and verify Green's theorem. (3 marks)

Good luck!!!